

ISSTA 2026 · OAKLAND, CA · **ACM SIGSOFT DISTINGUISHED PAPER AWARD**

Do Coverage and Mutation Scores of LLM-Generated Test Suites Correlate with Their Effectiveness?

(Replicability Study)

Junda Zhao¹ · Shurui Zhou² · Eldan Cohen¹

¹ Mechanical & Industrial Engineering ² Electrical & Computer Engineering



UNIVERSITY OF
TORONTO

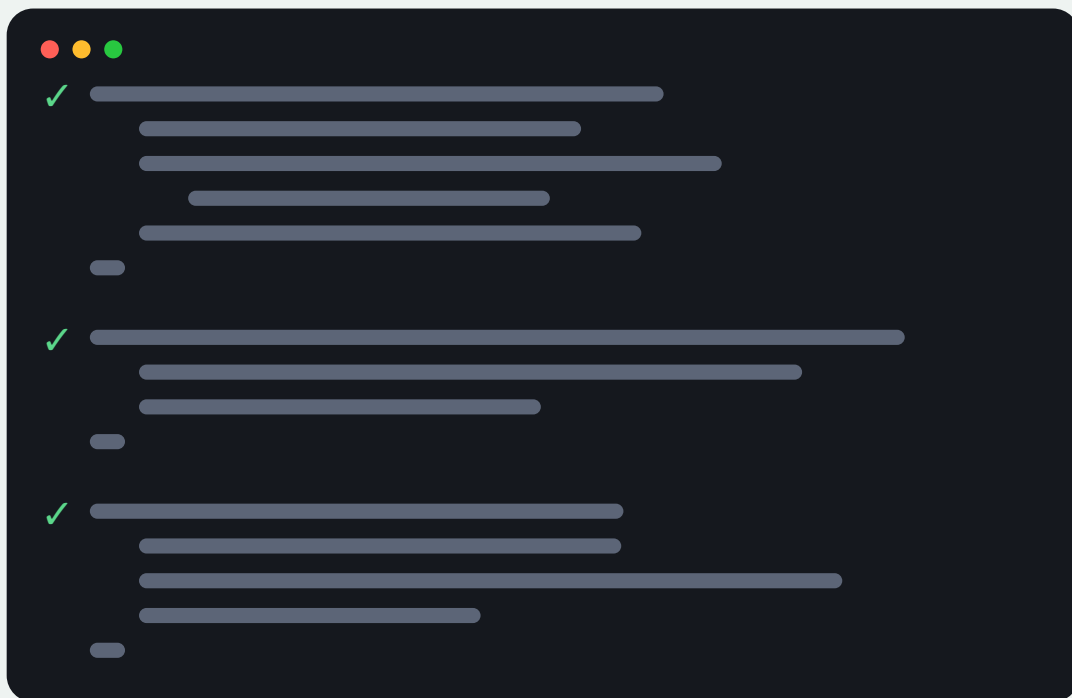


What makes a **good** test suite nowadays?

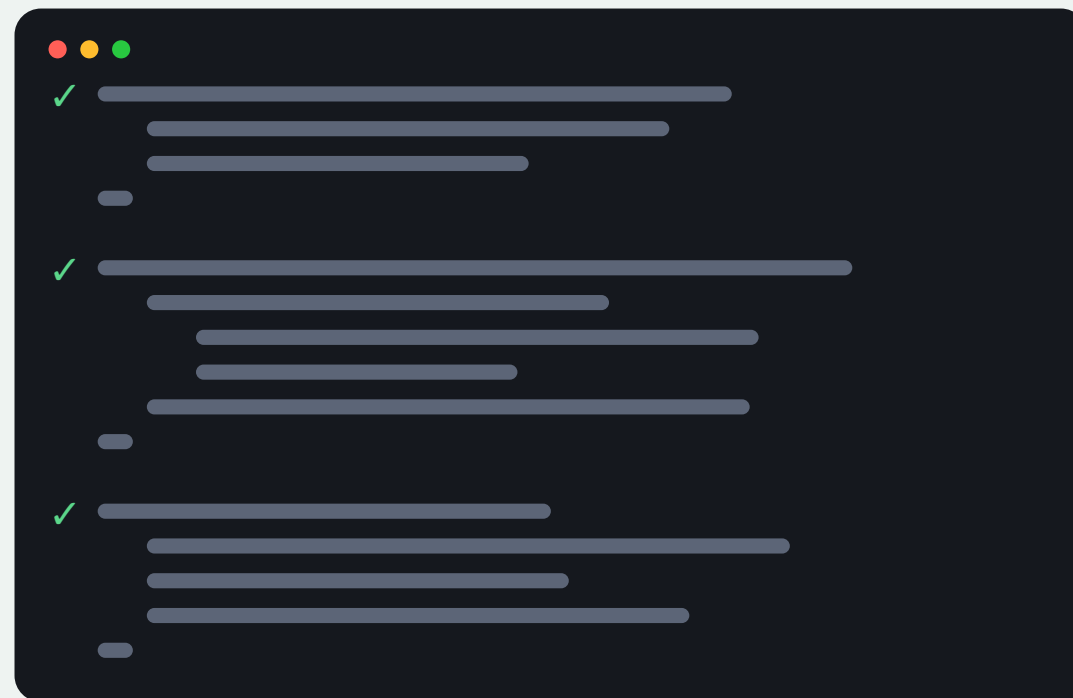
A HYPOTHETICAL

Two LLMs write test suites for the same code

LLM A



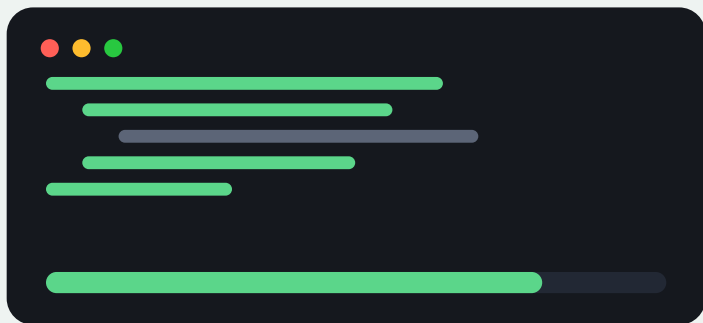
LLM B



How do we tell which one is **better**?

Tests are commonly evaluated by proxies

INDUSTRY



Coverage: one of the most important measures of test quality

Required or recommended by standards, e.g. DO-178C, ISO 26262

PRIOR STUDIES

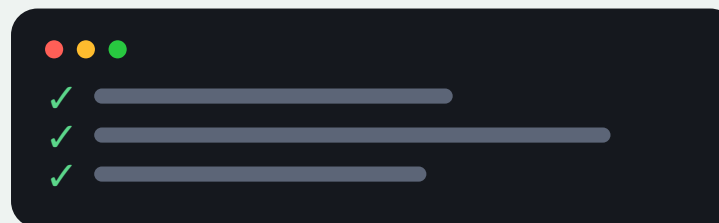


Coverage: among the most reported outcomes

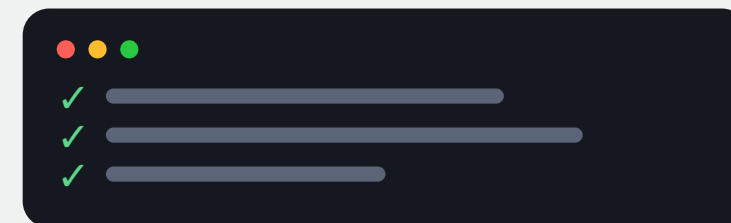
Mutation score: a proxy for bug detection

LLM A scores higher on both measures

LLM A

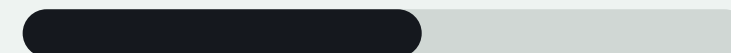
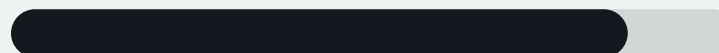


LLM B



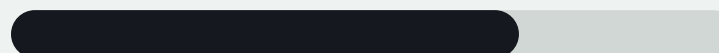
Coverage

Did the tests execute this code?



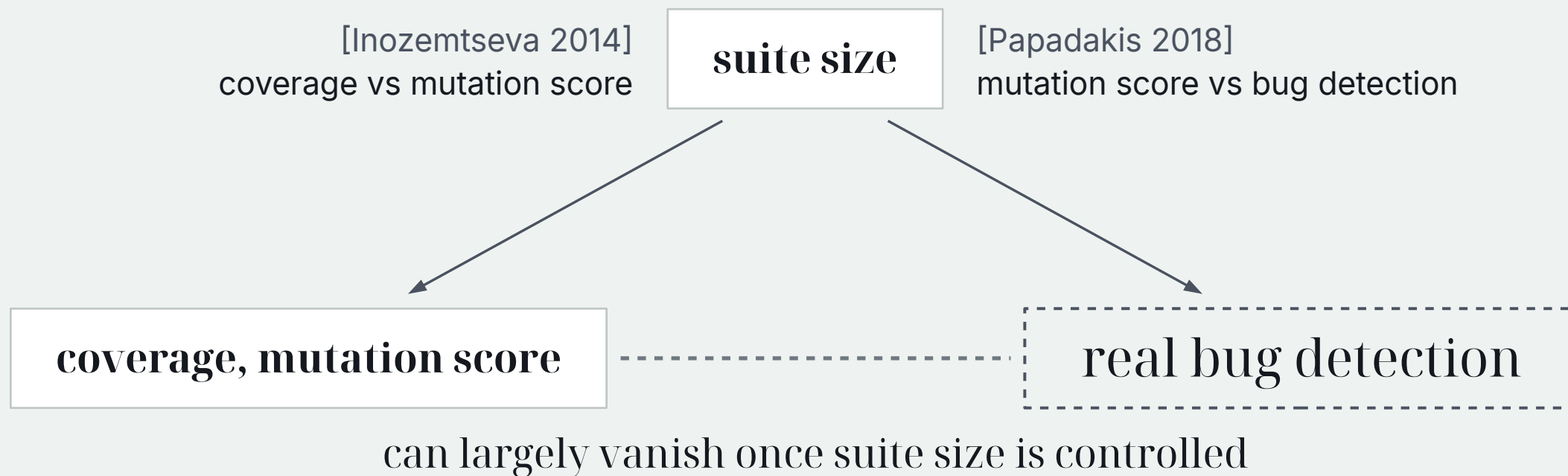
Mutation score

Would the tests notice
if this code changed?



Real bug detection?

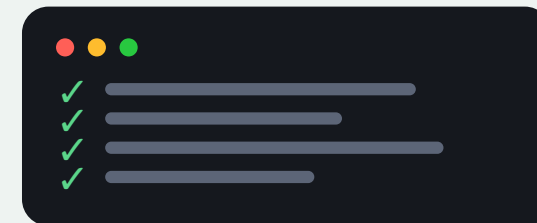
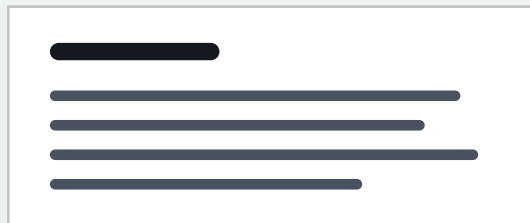
Correlated, but with a catch



If the proxies do not track bug detection,
evaluating through them **may mislead**

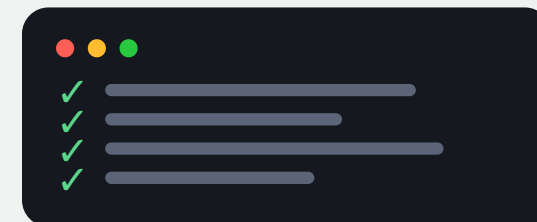
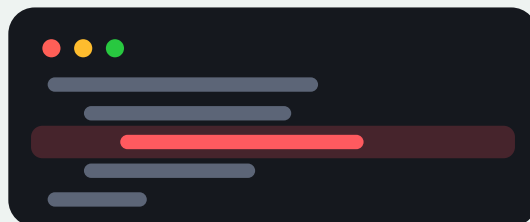
LLMs often write tests differently from humans

human-written



often grounded in developers' understanding
of the intended behavior

LLM-generated



code-under-test: the primary prompt input
may be buggy

Still common: coverage, mutation score
Real bug detection: rarely reported

Do the conclusions for human-written tests **carry over**?

We replicate both studies on LLM tests

8,268 101,123

suites

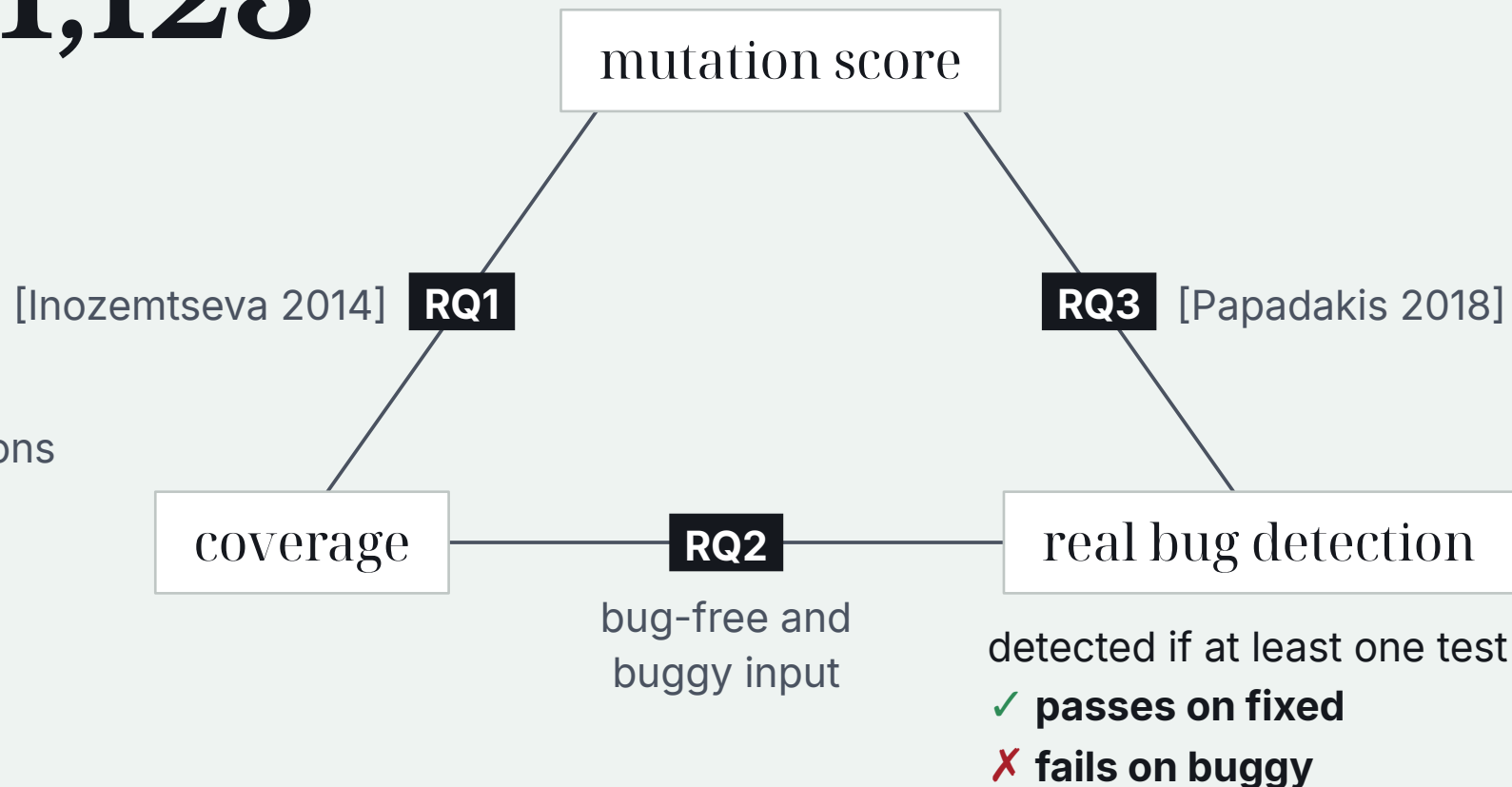
tests

318

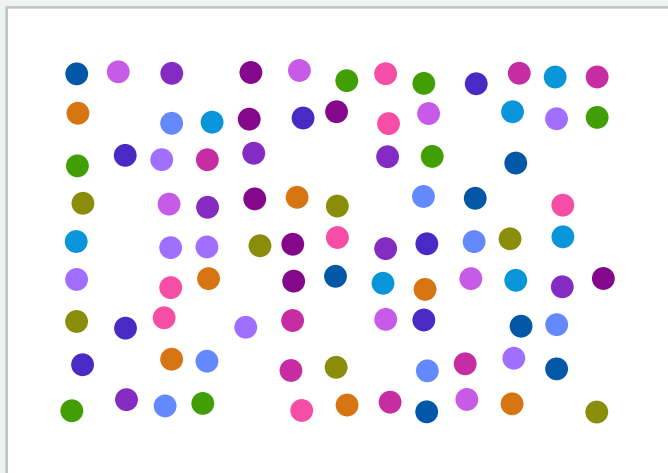
focal methods from Defects4J,
each with buggy and fixed versions

13

configurations of 11 LLMs

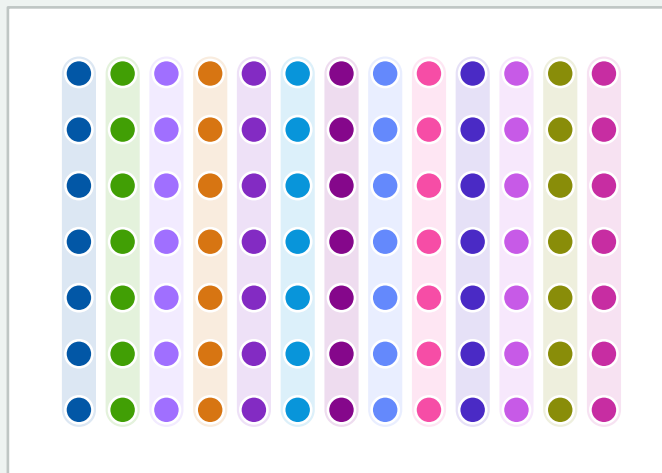


We analyze each correlation in three views



all models pooled

1,000 draws of 100 suites
per model



within a model

separately for each model



across models

each model becomes one point

Suite size fixed:

3, 5 or 10 tests
kept at random

Average: per-method ratio, then the mean

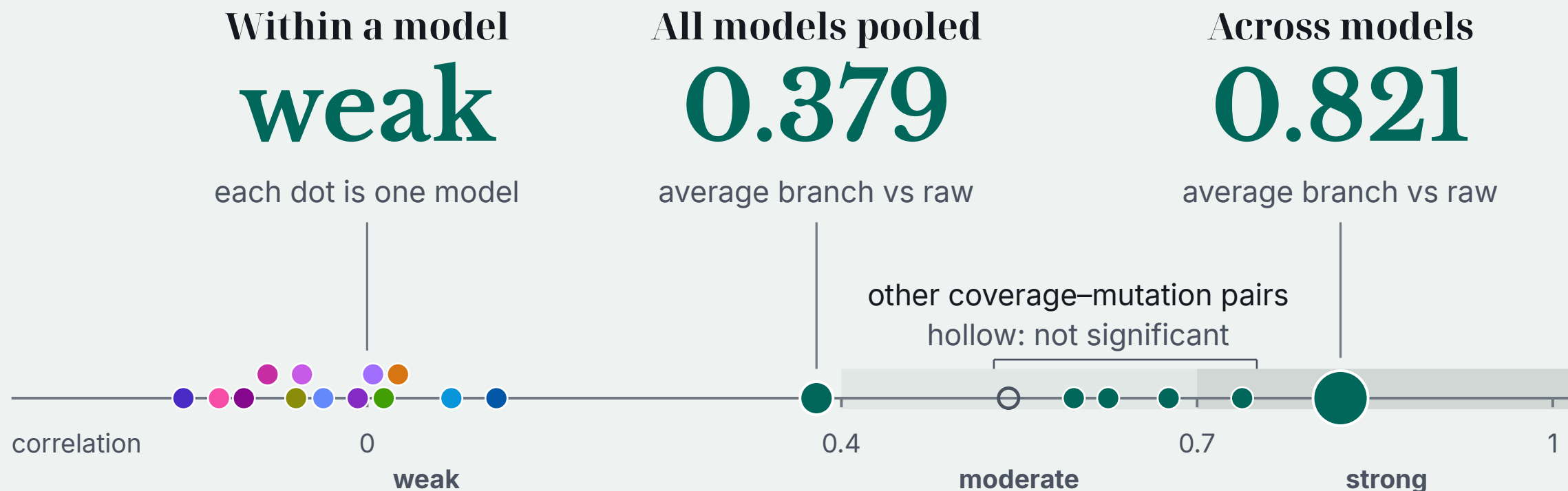
Accumulated: one ratio over all methods

Raw: killed over all mutants

Normalized: killed over covered mutants

So, do the conclusions
carry over?

Weak within, moderate to strong across



Coverage is informative for **comparing models**

Coverage types: related, not interchangeable

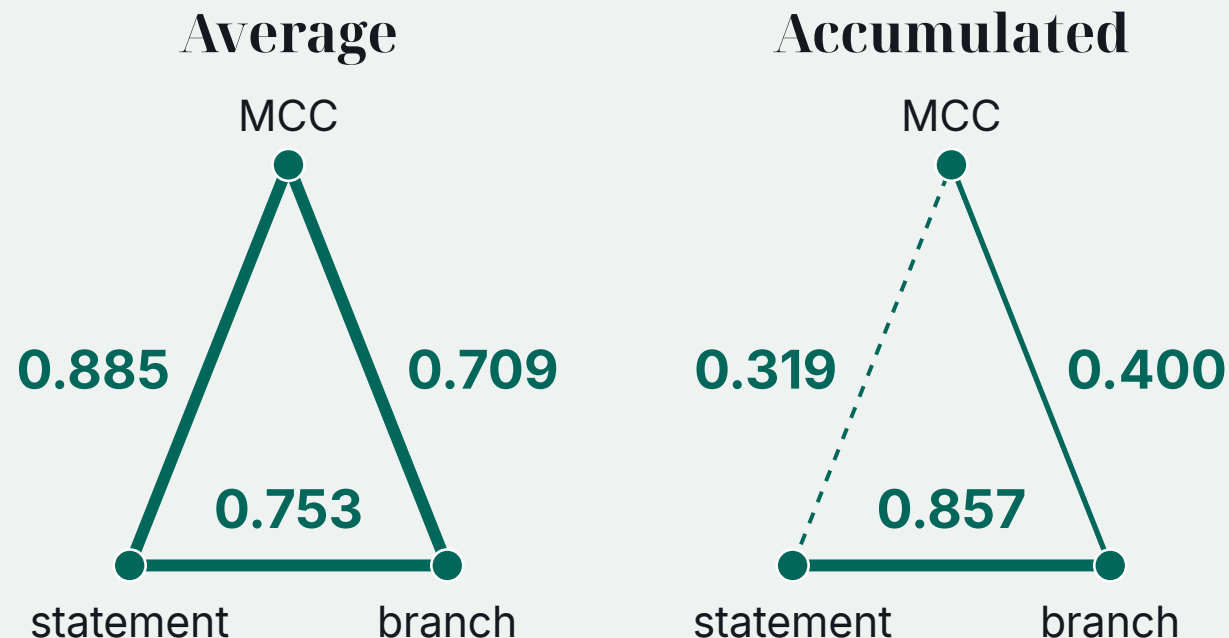
HUMAN-WRITTEN [Inozemtseva 2014]

very strong

all Pearson r above 0.95
between coverage types

Their conclusion: “use simpler coverage measures”, saving measurement effort

LLM-GENERATED · ALL MODELS POOLED



“Use simpler coverage measures” **does not necessarily hold** here

Bug detection: the same pattern, stronger

Within a model
below 0.2

typically



Across models

0.861

average branch

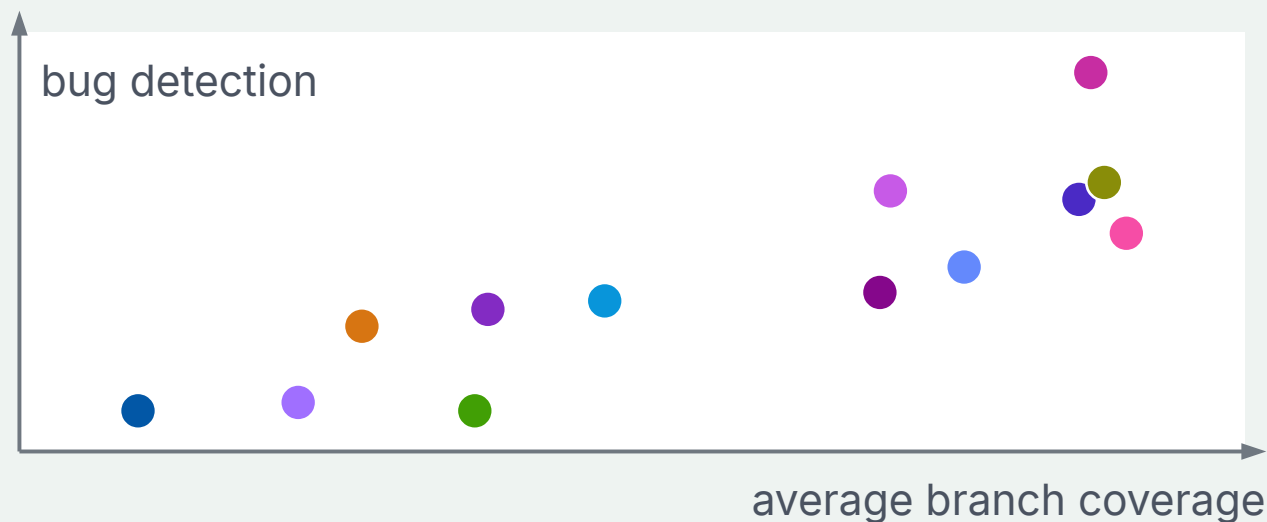
moderate

0.617

statement

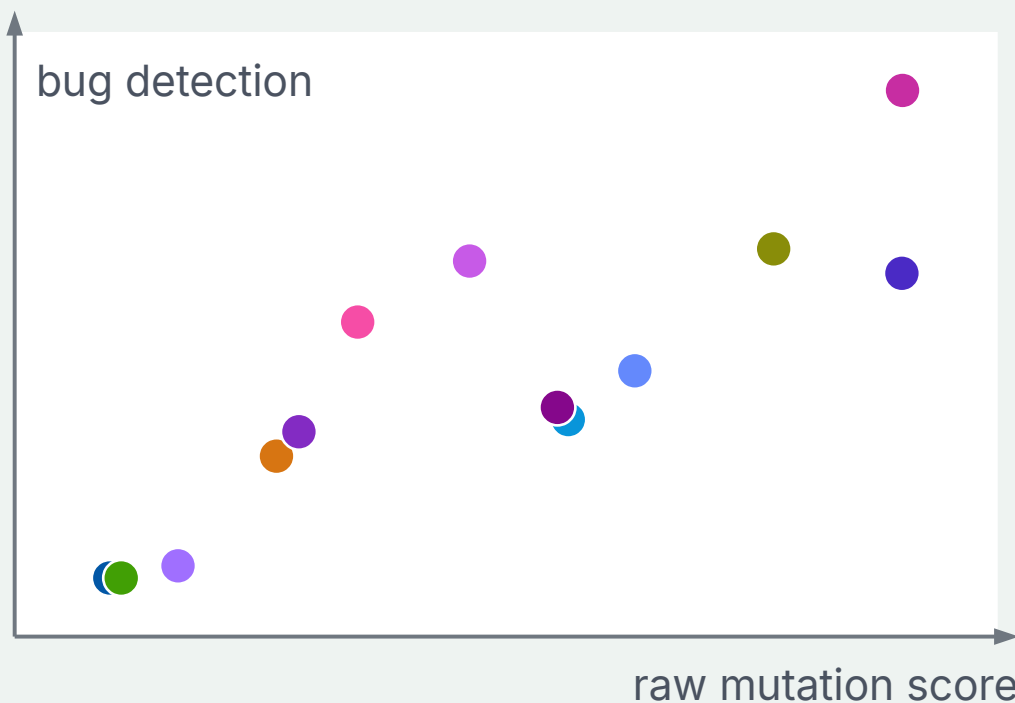
0.573

MCC



Coverage can be indicative of bug detection **across models**

Raw mutation score: strong across models



AVERAGE AGGREGATION	RAW	NORMALIZED
All models pooled	0.475	0.511
Within a model	weak	weak
Across models	0.863	0.493

grey: not significant

Raw mutation score can be informative for **comparing models**

TEST SUITE SIZE

Suite size: only weakly correlated

HUMAN-WRITTEN

Both studies: a strong confounder

strong

positive correlation

with mutation score [Inozemtseva 2014]

LLM-GENERATED

weak

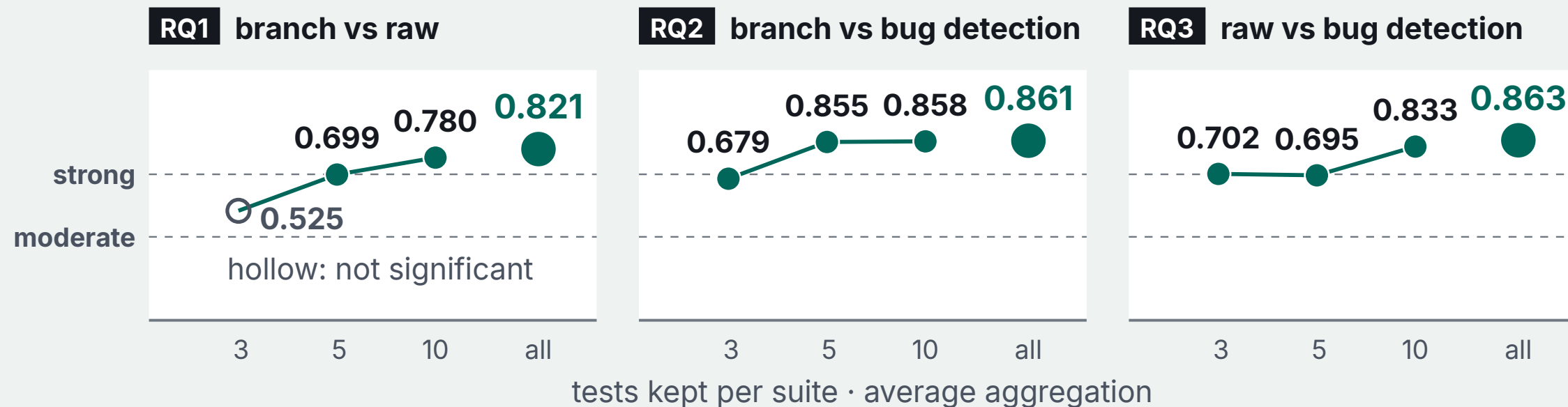
Mutation score: weak in every view,
variant and aggregation

Bug detection: weak in every view

Coverage: no strong relationship observed

Unlike prior work, suite size shows **no strong correlation**

Size fixed: qualitative strength largely retained



No strong confounding effect from suite size is observed

So coverage and
mutation score can
be indicative... right?

A test that asserts the bug

from the
bug-free method

```
Lang-14, a representative example
@Test
public void testEquals_stringBuilder() {
    assertTrue(StringUtils.equals("abc",
        new StringBuilder("abc")));
}
```

from the
buggy method

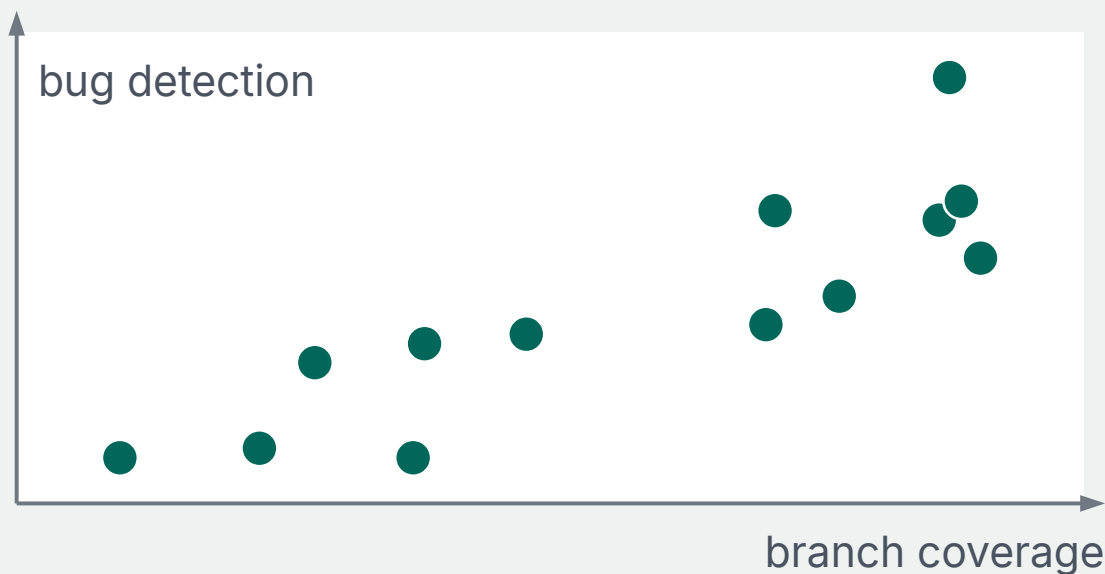
```
@Test
public void testEquals_stringBuilder() {
    assertFalse(StringUtils.equals("abc",
        new StringBuilder("abc")));
}
```

Tests from buggy code can **encode faulty behavior**

The strong correlation turns weak

BUG-FREE INPUT

0.861 across models · average branch



BUGGY INPUT

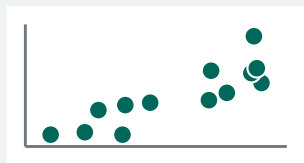
weak pooled, within and across models



Mutation score: **not applicable**

TAKEAWAYS

What did we learn that is new?



Bug-free input

both proxies can be indicative
across models, even with size fixed



Buggy input

the strong correlation turns weak;
measure bug detection directly

Coverage criteria: complementary signals;
report several, including MCC

Suite size: little evidence of a dominant confounder

PRIOR WORK · HUMAN-WRITTEN

can largely vanish once
suite size is controlled

"use simpler
coverage measures"

a strong confounder

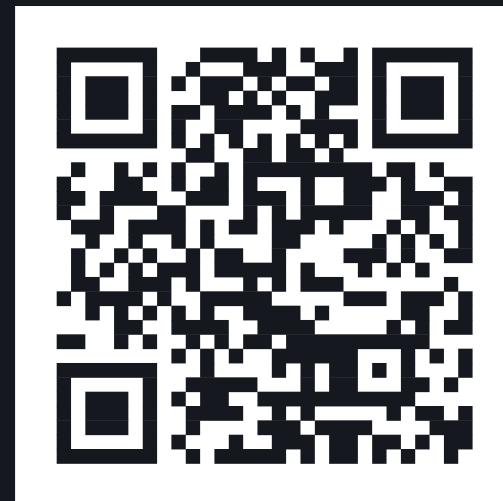
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Thank you

junda.zhao@mail.utoronto.ca

On the job market for postdoc positions

Happy to chat, open to questions.



Paper (arXiv)



Replication package



Recommended in industry

ISO 26262-6:2018 §9.4.4: structural coverage at unit level

Coverage criterion	ASIL A	ASIL B	ASIL C	ASIL D
Statement coverage	++	++	+	+
Branch coverage	+	++	++	++
MC/DC	+	+	+	++

++ highly recommended + recommended

Airborne software, Level A: MC/DC required [RTCA DO-178C]

MCC in the paper is the standard's MC/DC.

The reviewed evaluations

Study	Models	Coverage	Mutation	Bug detection	Code-under-test in prompt	Buggy code as input
Tufano et al. 2021	BART	●			●	
Schäfer et al. 2024	GPT-3.5	●			●	
Guilherme et al. 2023	GPT-3.5-turbo	●	●		●	
Rao et al. 2023	GPT-2	●			●	
Tang et al. 2024	GPT-3	●		●	●	
Siddiq et al. 2024	Codex, StarCoder, GPT-3.5-turbo	●			●	
Yuan et al. 2024	GPT-3.5-turbo	●			●	
Bhatia et al. 2024	GPT-3.5	●			●	
Yang et al. 2024	6 models	●		●	●	
Zhang et al. 2024	CodeLlama, GPT-3.5, GPT-4	●	●		●	
Lops et al. 2025	GPT-3.5-turbo, GPT-4	●	●		●	
Shang et al. 2025	37 models				●	
Jain et al. 2025	10 models	●	●		●	
Wang et al. 2025	17 models	●			●	
Abdullin et al. 2025	4 models	●	●	●	●	
Huang et al. 2025	11 models	●		●	●	●

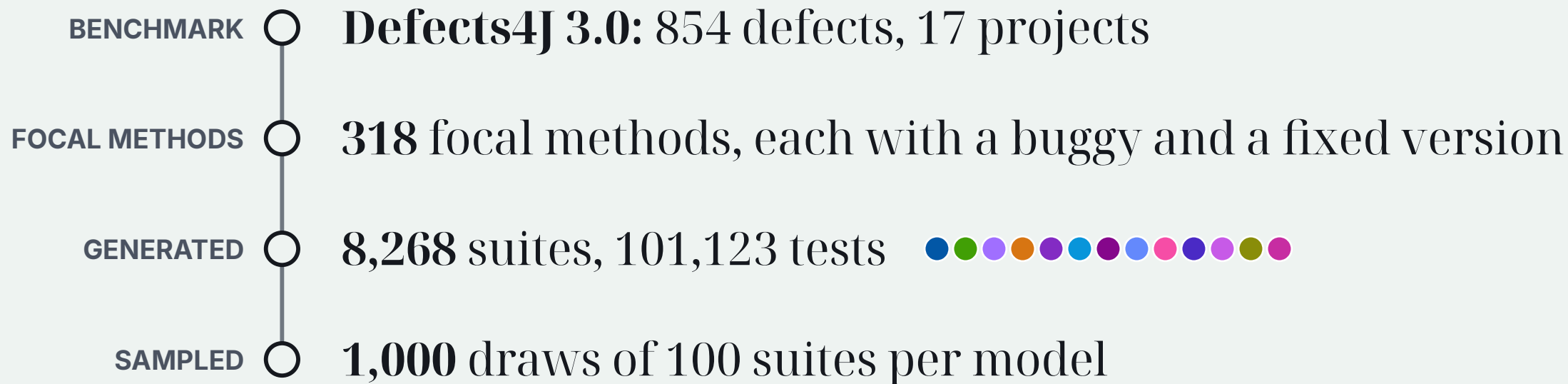
Mutation score: 5 studies · **Real-bug detection:** 4 studies · **Buggy code as input:** 1 study (Huang et al.)

The 13 configurations

Configuration	Developer	Type	Open weights
● Gemini 2.5 Flash	Google	Hybrid	No
● Gemini 2.5 Flash (Reason)	Google	Hybrid	No
● DeepSeek-V3	DeepSeek	Base	Yes
● Grok-3	xAI	Base	No
● Qwen3-Coder-Plus	Alibaba	Base	Yes
● GPT-4.1	OpenAI	Base	No
● Qwen3-Plus	Alibaba	Reasoning	Yes
● DeepSeek-R1	DeepSeek	Reasoning	Yes
● Claude 4 Sonnet (Reason)	Anthropic	Hybrid	No
● GPT-O4-mini	OpenAI	Reasoning	No
● Claude 4 Sonnet	Anthropic	Hybrid	No
● Grok-4	xAI	Reasoning	No
● Gemini 2.5 Pro	Google	Reasoning	No

11 LLMs from 6 developers, 13 configurations

Benchmark and sampling



Suite size fixed: $k = 3, 5$ or 10 tests kept at random, from suites with at least k tests

Temperature 0

Definitions

Coverage:	statement, branch and modified condition coverage (MCC), by CodeCover
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Raw mutation score:	killed mutants over all mutants of the focal method
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Normalized mutation score:	killed mutants over the mutants the suite covers
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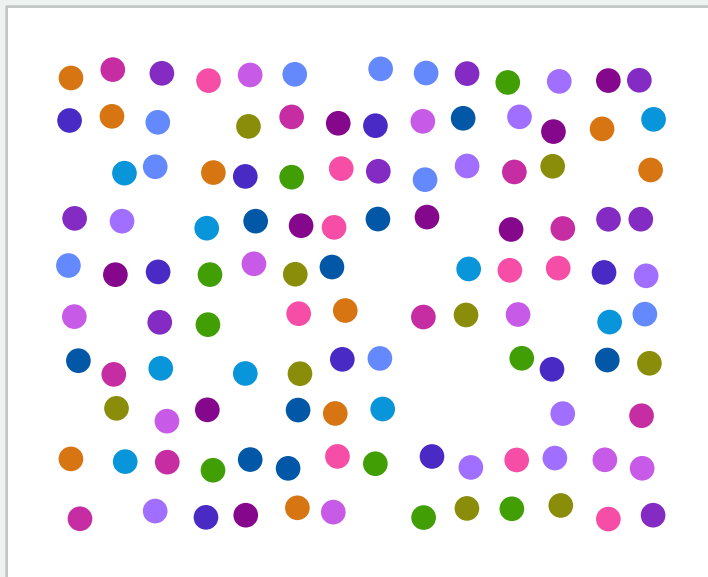
Average aggregation:	the per-method ratio averaged over the methods
-----------------------------	--

Bug detection ratio:	the proportion of buggy focal methods detected
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Guilford scale:	weak below 0.4, moderate 0.4 to 0.7, strong 0.7 to 0.9, very strong above 0.9
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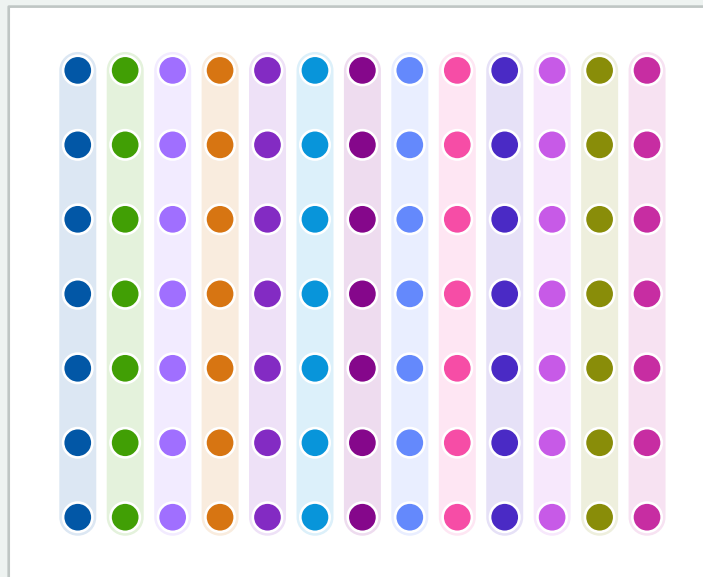


The three views, in the paper's words



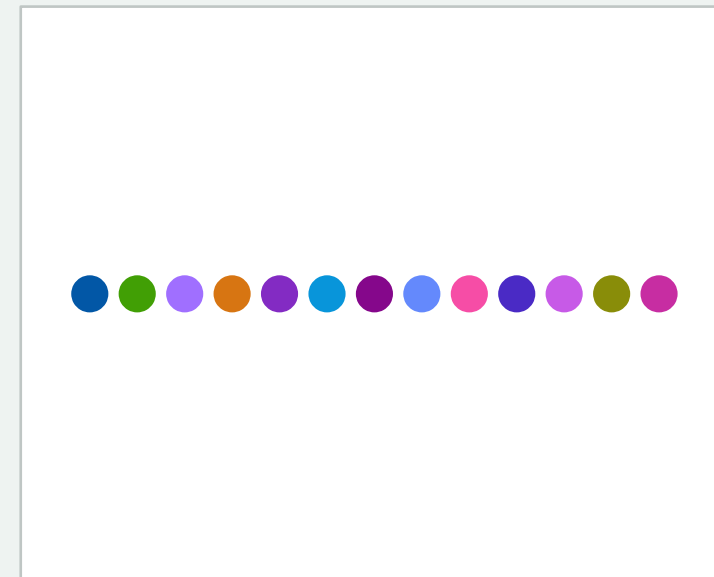
Combined

all models pooled



Intra-model

within a model



Inter-model

across models

Inter-model results are read “primarily through statistically significant coefficients”.

RQ1: coverage vs mutation score

			All models pooled		Across models	
Coverage	Mutation score		Average	Accumulated	Average	Accumulated
Statement	Raw	Pearson r	0.356	0.206	0.676 (0.011)	0.383 (0.197)
		Kendall τ	0.250	0.162	0.462 (0.030)	0.410 (0.057)
	Normalized	Pearson r	0.443	-0.009	0.738 (0.004)	-0.047 (0.880)
		Kendall τ	0.272	-0.004	0.538 (0.010)	-0.051 (0.858)
Branch	Raw	Pearson r	0.379	0.142	0.821 (0.001)	0.246 (0.419)
		Kendall τ	0.270	0.109	0.641 (0.002)	0.154 (0.510)
	Normalized	Pearson r	0.348	0.032	0.596 (0.032)	0.035 (0.909)
		Kendall τ	0.238	0.022	0.410 (0.057)	0.051 (0.858)
MCC	Raw	Pearson r	0.267	0.118	0.541 (0.057)	0.289 (0.339)
		Kendall τ	0.186	0.092	0.385 (0.076)	0.308 (0.163)
	Normalized	Pearson r	0.339	-0.087	0.625 (0.022)	-0.124 (0.686)
		Kendall τ	0.218	-0.064	0.462 (0.030)	-0.051 (0.858)

In parentheses: p. In grey: not significant at $p < 0.05$.

RQ1 within each model, as published

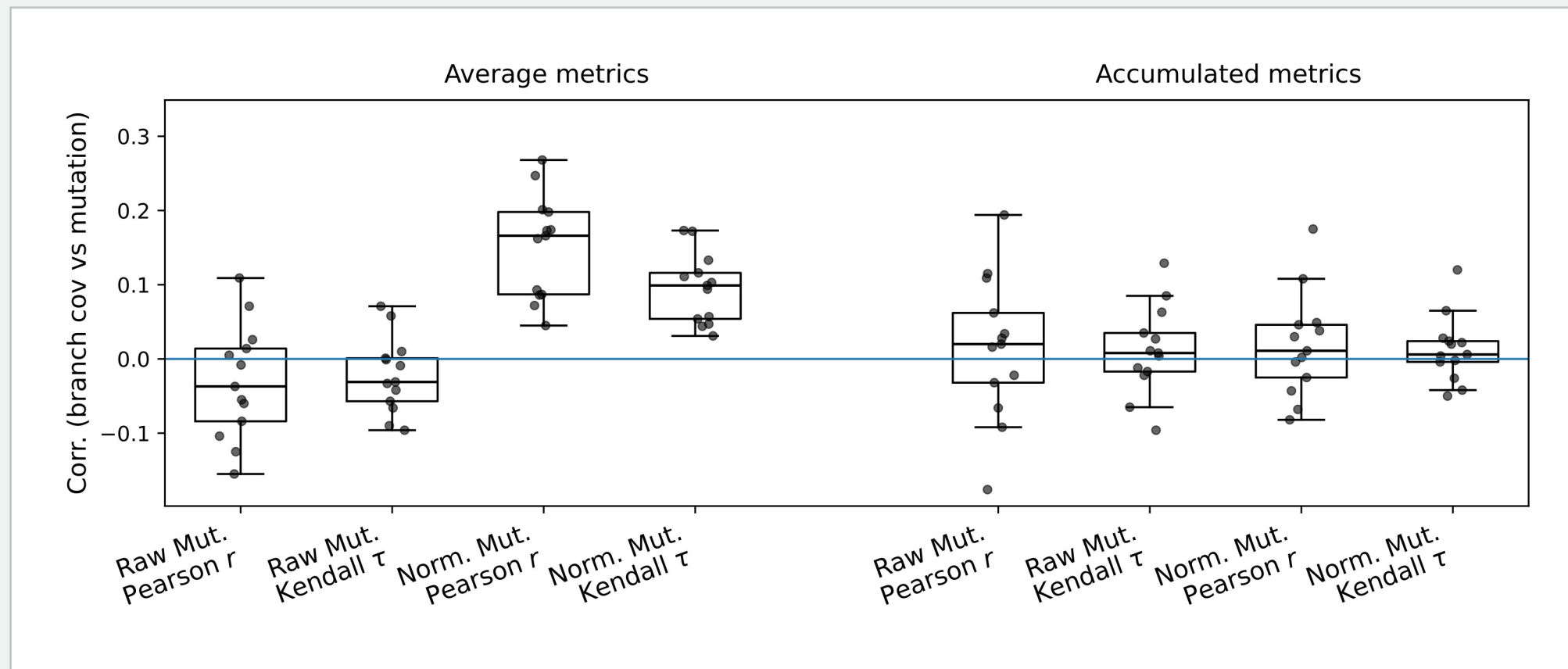


Figure 3b: Branch coverage vs. mutation score.

RQ1 with suite size fixed

Across models			Average			Accumulated		
Coverage	Mutation score		3 tests	5 tests	10 tests	3 tests	5 tests	10 tests
Statement	Raw	Pearson r	0.190 (0.534)	0.476 (0.100)	0.632 (0.021)	−0.144 (0.639)	0.489 (0.090)	0.140 (0.648)
		Kendall τ	0.103 (0.675)	0.385 (0.076)	0.590 (0.004)	−0.103 (0.675)	0.282 (0.204)	0.179 (0.435)
	Normalized	Pearson r	0.642 (0.018)	0.462 (0.112)	0.688 (0.009)	−0.251 (0.409)	0.009 (0.977)	−0.263 (0.386)
		Kendall τ	0.410 (0.057)	0.282 (0.204)	0.513 (0.015)	−0.128 (0.590)	0.103 (0.675)	−0.282 (0.204)
Branch	Raw	Pearson r	0.525 (0.066)	0.699 (0.008)	0.780 (0.002)	−0.102 (0.739)	−0.001 (0.997)	0.054 (0.860)
		Kendall τ	0.282 (0.204)	0.513 (0.015)	0.692 (0.001)	−0.026 (0.952)	−0.154 (0.510)	−0.154 (0.510)
	Normalized	Pearson r	0.579 (0.038)	0.323 (0.282)	0.528 (0.064)	0.036 (0.907)	0.165 (0.589)	0.072 (0.816)
		Kendall τ	0.487 (0.022)	0.256 (0.252)	0.410 (0.057)	0.000 (1.000)	0.179 (0.435)	−0.103 (0.675)
MCC	Raw	Pearson r	0.035 (0.910)	0.476 (0.100)	0.550 (0.052)	0.156 (0.610)	0.015 (0.962)	0.095 (0.759)
		Kendall τ	0.000 (1.000)	0.410 (0.057)	0.462 (0.030)	0.051 (0.858)	−0.077 (0.765)	0.077 (0.765)
	Normalized	Pearson r	0.669 (0.012)	0.526 (0.065)	0.652 (0.016)	0.127 (0.680)	−0.283 (0.350)	−0.492 (0.088)
		Kendall τ	0.462 (0.030)	0.410 (0.057)	0.487 (0.022)	0.128 (0.590)	−0.154 (0.510)	−0.333 (0.129)

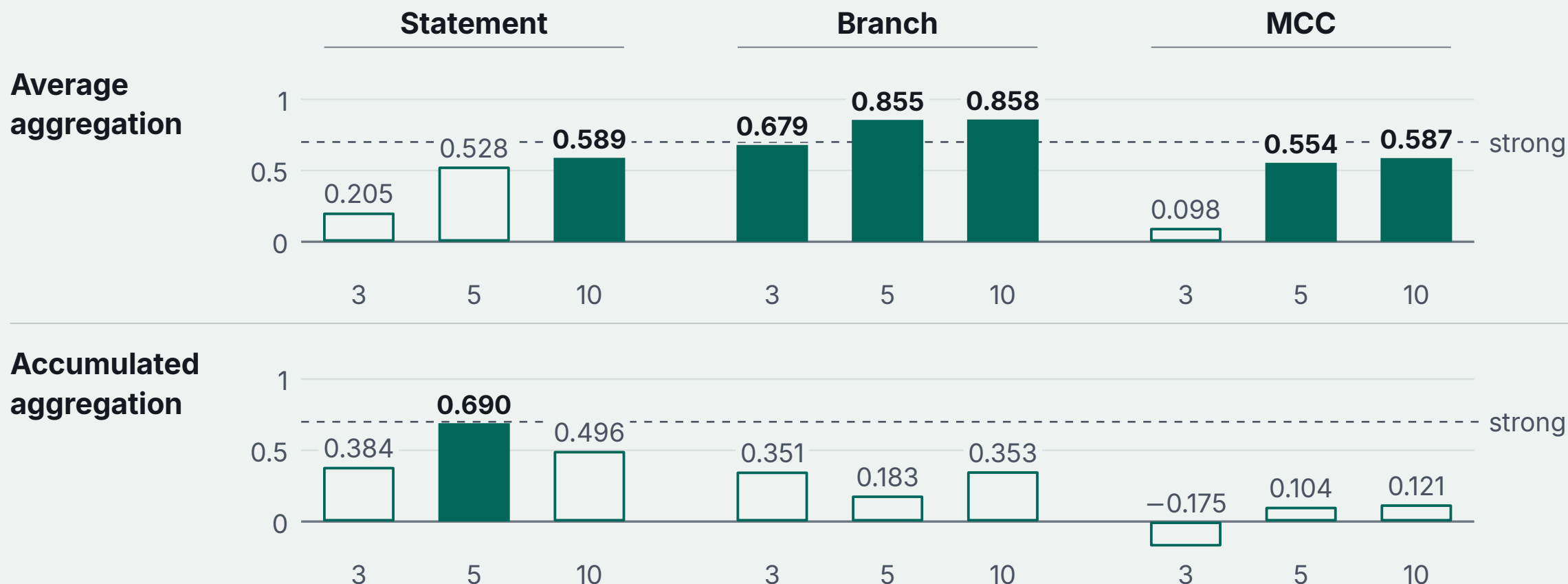
In parentheses: p. In grey: not significant at $p < 0.05$.

RQ2: coverage vs real bug detection

		All models pooled		Across models	
Coverage		Average	Accumulated	Average	Accumulated
Statement	Pearson r	0.368	0.473	0.617 (0.025)	0.619 (0.024)
	Kendall τ	0.250	0.336	0.477 (0.024)	0.555 (0.0086)
Branch	Pearson r	0.385	0.481	0.861 (1.6×10^{-4})	0.542 (0.056)
	Kendall τ	0.266	0.337	0.761 (3.1×10^{-4})	0.400 (0.058)
MCC	Pearson r	0.379	0.190	0.573 (0.041)	0.504 (0.079)
	Kendall τ	0.261	0.127	0.503 (0.017)	0.452 (0.032)

In parentheses: p. In grey: not significant at $p < 0.05$.

RQ2 with suite size fixed, all criteria



Across models; bars at 3, 5 and 10 tests per suite. Hollow: not significant at $p < 0.05$ (the printed 0.050 counts as significant).

RQ3: mutation score vs real bug detection

Aggregation	Mutation score		All models pooled size ignored	Across models			
				size ignored	3 tests	5 tests	10 tests
Average	Raw	Pearson r	0.475	0.863 (1.44×10^{-4})	0.702 (0.007)	0.695 (0.008)	0.833 (4.05×10^{-4})
		Kendall τ	0.320	0.761 (3.11×10^{-4})	0.510 (0.017)	0.510 (0.017)	0.658 (1.83×10^{-3})
	Normalized	Pearson r	0.511	0.493 (0.087)	0.354 (0.236)	0.386 (0.192)	0.532 (0.061)
		Kendall τ	0.373	0.374 (0.076)	0.222 (0.297)	0.275 (0.197)	0.426 (0.044)
Accumulated	Raw	Pearson r	0.146	0.622 (0.023)	0.585 (0.036)	0.533 (0.061)	0.600 (0.030)
		Kendall τ	0.110	0.529 (0.012)	0.405 (0.057)	0.301 (0.158)	0.477 (0.024)
	Normalized	Pearson r	0.196	-0.077 (0.803)	0.186 (0.542)	-0.186 (0.542)	-0.085 (0.783)
		Kendall τ	0.123	-0.039 (0.855)	0.013 (0.951)	-0.118 (0.580)	0.013 (0.951)

In parentheses: p. In grey: not significant at $p < 0.05$.

Suite size, all views

SUITE SIZE VS MUTATION SCORE

Aggregation	Mutation score		All models pooled	Across models
Average	Raw	Pearson r	0.031	0.185 (0.546)
		Kendall τ	0.034	0.205 (0.367)
	Normalized	Pearson r	0.098	0.093 (0.761)
		Kendall τ	0.066	0.026 (0.952)
Accumulated	Raw	Pearson r	0.029	0.259 (0.393)
		Kendall τ	0.017	0.231 (0.306)
	Normalized	Pearson r	−0.112	−0.347 (0.245)
		Kendall τ	−0.047	−0.231 (0.306)

In parentheses: p. In grey: not significant at $p < 0.05$.

Size vs bug detection, across models:

around 0.4, just below the moderate threshold, not significant

Why suite size behaves differently: the paper's words

The weaker confounding effect “likely stems from how varying-sized test suites are constructed”.

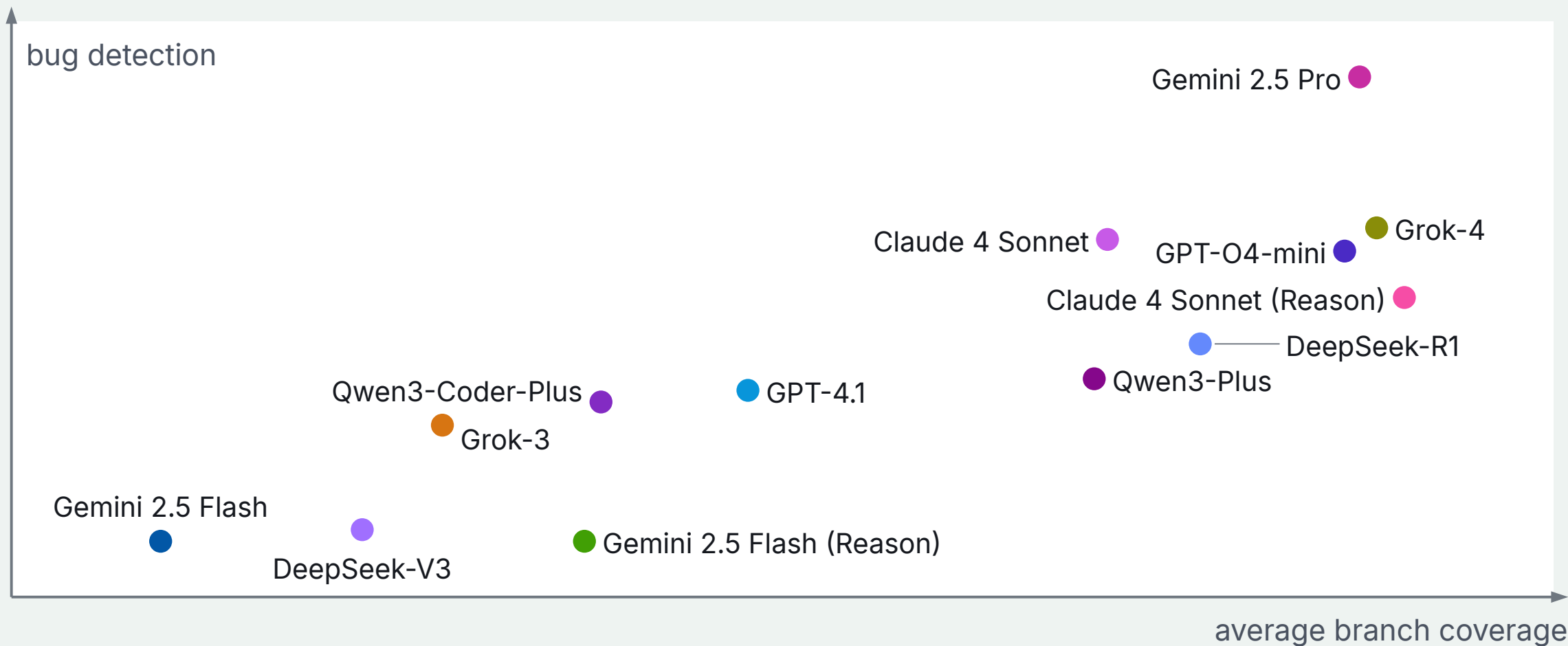
LLM-GENERATED

“In our setting, each test suite is generated by the LLM for a single focal method, so differences in suite size likely reflect how many tests are needed to exercise the set of behaviors required to validate that method's expected functionality.”

HUMAN-WRITTEN

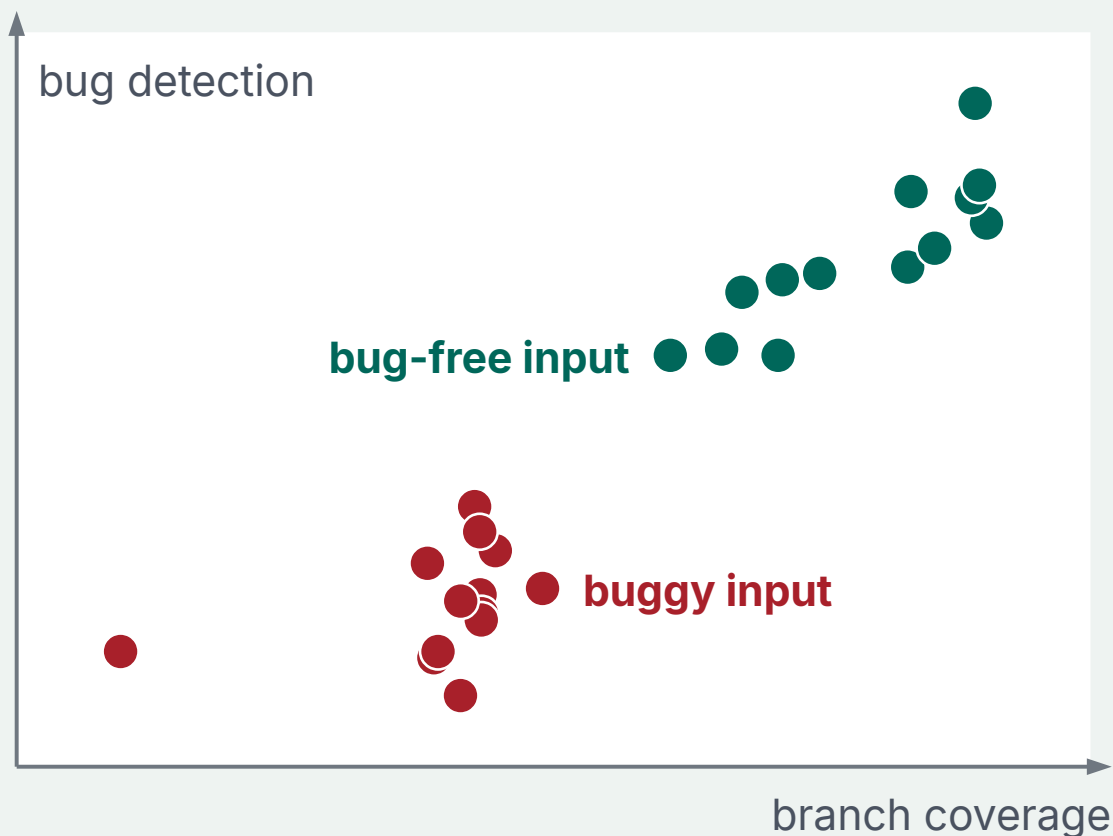
“In contrast, prior studies construct suites by sampling a specified number of tests from a project-wide test pool.”

The 13 points, named



Buggy input: what the paper reports

positions from the replication package, both axes shared



“The correlations are weak under the combined, intra-model, and inter-model views.”

Mutation testing “presupposes a passing (‘green’) test suite”: “Mutation analysis is therefore not applicable in this setting.”

Lang-14 in full

● ● ● Bug-free method

```
public static boolean equals(
    CharSequence cs1, CharSequence cs2) {
    if (cs1 == cs2)          return true;
    if (cs1 == null
        || cs2 == null)      return false;
    if (cs1 instanceof String
        && cs2 instanceof String)
        return cs1.equals(cs2);
    return CharSequenceUtils.regionMatches(
        cs1, false, 0, cs2, 0,
        Math.max(cs1.length(), cs2.length()));
}
```

● ● ● Buggy method

```
public static boolean equals(
    CharSequence cs1, CharSequence cs2) {
    if (cs1 == cs2)          return true;
    if (cs1 == null
        || cs2 == null)      return false;
    return cs1.equals(cs2);
}
```

Coverage criteria and guidance

COVERAGE VS COVERAGE, ALL MODELS POOLED

Suite size	Coverage pair	Average		Accumulated	
		r	τ	r	τ
All	Statement vs Branch	0.753	0.553	0.857	0.675
	Statement vs MCC	0.885	0.694	0.319	0.222
	Branch vs MCC	0.709	0.510	0.400	0.271
3 tests	Statement vs Branch	0.663	0.468	0.811	0.614
	Statement vs MCC	0.789	0.572	−0.003	0.005
	Branch vs MCC	0.458	0.303	−0.059	−0.037
5 tests	Statement vs Branch	0.690	0.496	0.699	0.506
	Statement vs MCC	0.839	0.639	0.209	0.141
	Branch vs MCC	0.605	0.413	−0.021	−0.008
10 tests	Statement vs Branch	0.715	0.517	0.772	0.566
	Statement vs MCC	0.878	0.683	0.189	0.130
	Branch vs MCC	0.665	0.467	0.034	0.027

In grey: not significant at $p < 0.05$.

Results suggest different coverage types “are not interchangeable”.

Correctness measures are “better viewed as measures of cost-effectiveness”.

Threats to validity

External

Java and Defects4J 3.0 only

Internal

Inter-model view: one point per model,
read primarily through significant coefficients

Construct

Temperature 0; some nondeterminism may remain

Data contamination: "we cannot rule this out",
but we observe "substantial differences between tests generated from
buggy versus fixed versions"

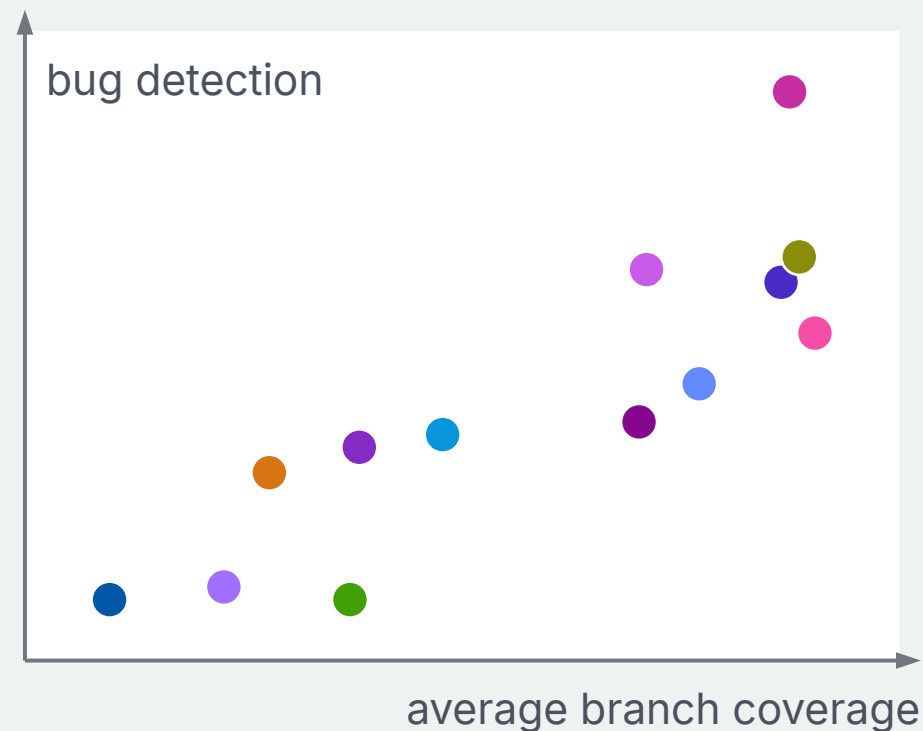
Where the dots come from

Positions from the public replication package,
aggregated as its notebooks do

They reproduce the paper's across-model
coefficients and its pooled RQ2 coefficient

Not drawn: the pooled RQ1 and suite-size
clouds, which do not reproduce

The axes carry no numbers: average coverage
counts branch-free methods as zero



What prior work concluded, in our paper's summary

Prior work, human-written tests	LLM-generated tests, this study	Evidence
Inozemtseva et al.: the correlation can largely disappear once test suite size is controlled; Papadakis et al. report analogous findings	Coverage and mutation can still be indicative of bug detection effectiveness, even when test suite size is controlled, but only in certain settings	Across models, suite size fixed: branch coverage vs bug detection 0.855 (5 tests), 0.858 (10 tests)
Inozemtseva et al.: "it may be in the developer's best interest to use simpler coverage measures"	Different coverage criteria provide complementary signals	All models pooled, accumulated: MCC vs statement 0.319, MCC vs branch 0.400
Both studies argue that test suite size is a strong confounder	Little evidence that test suite size is a dominant confounder	All models pooled: suite size vs raw mutation score 0.031